Bridge Material Type and Structural Condition

Build things that hold up, literally and statistically.

Gabriel Charris
Mechanical Engineering
University of South Florida

Connect on LinkedIn

Introduction

Steel corrodes and fatigues; prestressed concrete resists cracking. Does that show up in Florida's bridge ratings?

Florida's 13,070 inventoried bridges give a large sample to test whether material relates to condition, and whether Poor-rated bridges concentrate where salt air is assumed to do the most damage: the coast.

RESEARCH QUESTION

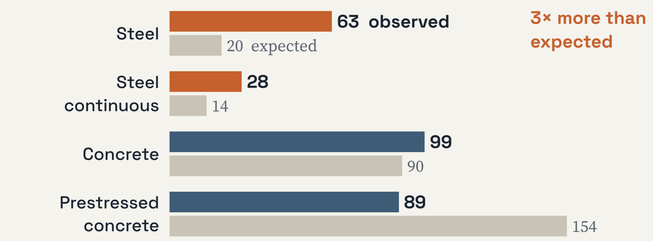
Is a Florida bridge's main material associated with its condition rating, and where are the Poor-rated steel bridges?

Methods

- **Data.** 2025 FHWA National Bridge Inventory: 13,070 Florida bridges. 11,680 in the four material classes were analyzed.
- **Material.** Steel, steel continuous, concrete, prestressed.
- **Condition.** FHWA Good / Fair / Poor, from the lowest component.
- **Hypotheses.** H_0: material and condition are independent. H_1: they are associated.
- **Test.** Chi-square test of independence on the 4 × 3 material-by-condition table (df = 6, α = 0.05), with Cramér's V for effect size.
- **Geography.** Northwest Florida = west of 83°W (Panhandle and Big Bend). Poor share of steel bridges compared on each side of that line.

Poor-rated bridges: observed vs. expected

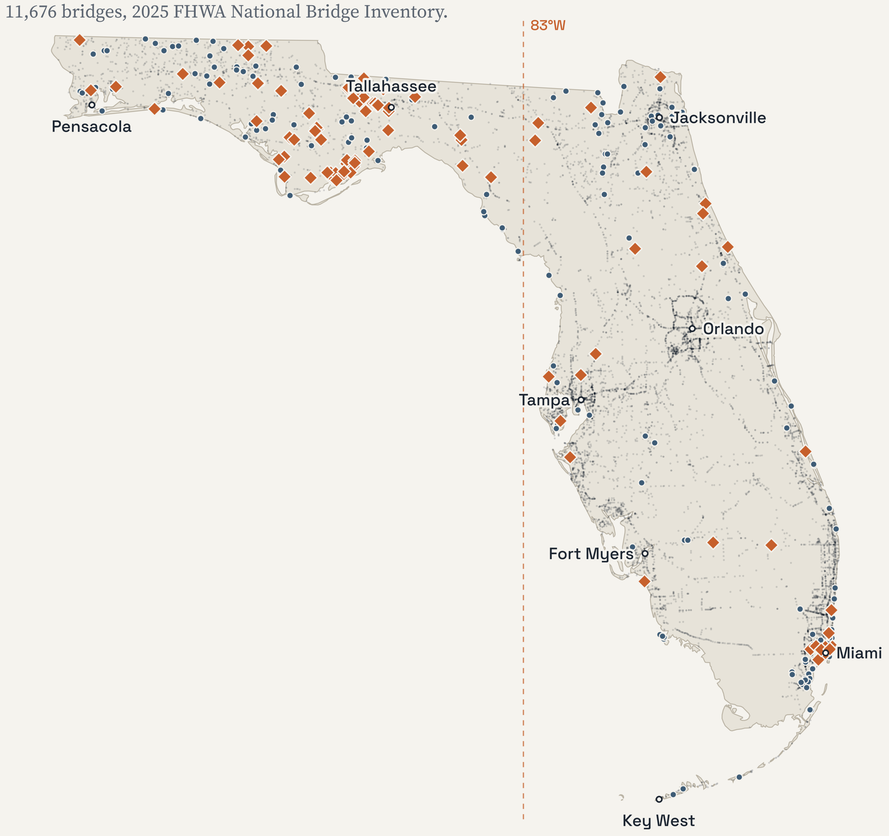
Expected = the count if every material had Florida's overall Poor rate (2.4%).



Where Florida's Poor-rated bridges are

Poor-rated steel bridges cluster in Northwest Florida, not along the coast.

11,676 bridges, 2025 FHWA National Bridge Inventory.



1 in 5

steel bridges in Northwest Florida is rated Poor.

Elsewhere in Florida: **1 in 40.**

Each dot is one bridge

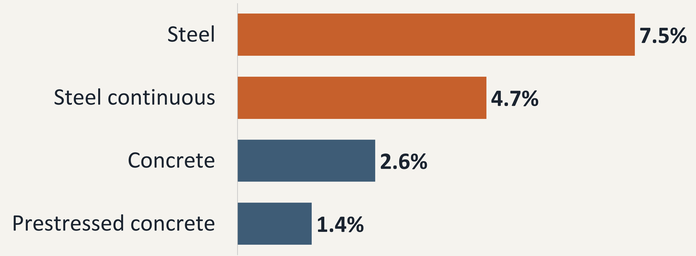
- Steel, rated Poor (91)
- Concrete, rated Poor (188)
- All other bridges (11,397)

Northwest Florida = west of 83°W (Panhandle and Big Bend): 62 of 281 steel bridges rated Poor. Rest of Florida: 29 of 1,164.
Steel includes simple and continuous spans. Concrete includes plain and prestressed.
Map shows 11,676 of 11,680 analyzed bridges (4 without valid coordinates not shown).

Results

Share of bridges rated Poor, by main material

n = 11,680 (steel 845; steel continuous 602; concrete 3,779; prestressed 6,454)



CHI-SQUARE TEST OF INDEPENDENCE

$\chi^2(6) = 485.26$

$p < 0.001$ · Cramér's V = 0.14

Material and condition are related; the effect size is small.

Discussion

- **Steel bridges are rated Poor 5× as often** as prestressed concrete (7.5% vs. 1.4%).
- **The steel gap is regional.** 22% of steel bridges in Northwest Florida are Poor, against 2.5% in the rest of the state.
- **The coast does not explain it.** Steel bridges within 2 km of the coast are not rated Poor more often than inland ones (5.0% vs. 6.7%, not significant).
- **Possible reasons:** Poor-rated steel bridges are older (median built 1965 vs. 1991), and in Northwest Florida, 28.5% of locally owned steel bridges are Poor vs. 6.2% of state-owned ones.
- **Implication:** inspect and repair older steel bridges in the Panhandle and Big Bend first.

Limitations

- Cross-sectional data show association, not cause. Age and ownership were compared, not modeled.
- With n = 11,680, small gaps reach significance; V = 0.14 is the better guide to size.

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Data: Federal Highway Administration, National Bridge Inventory (2025).